

Joshua Sternadel

Kansas City, MO | [Contact via LinkedIn](#)

jsternadel.github.io/portfolio/

Principal Systems & Software Engineer

Core Competencies: Full-Stack Architecture (Angular/Node/PHP) | Infrastructure Automation (Ansible, AAP) | Resilient Systems Design

Enterprise Scale: 13+ Years in Systems Engineering | Technical Lead Across Fortune 500 Telecom Mergers

U.S. Patent (#10,366,069) — Systems and Methods for Database Management and Administration (Co-Author)

Summary

Principal Systems and Software Engineer with 13+ years of experience designing enterprise platforms across telecommunications, infrastructure automation, distributed systems, and full-stack application architecture. Recognized for stabilizing mission-critical environments, leading large-scale migration initiatives, and building automation platforms that reduce operational complexity while improving reliability and developer productivity.

Key Achievements

- **U.S. Patent Author (#10,366,069):** Co-authored a patent for systems and methods for database management and administration, including a custom SQL natural sorting algorithm.
- **Mission-critical during the Sprint/T-Mobile merger:** Identified as essential infrastructure staff to stabilize systems and support architecture integration during the transition.
- **Custom data framework:** Built a PHP framework using reflection and annotations to unify SQL, NoSQL, JSON, and XML behind a single Entity Manager, reducing boilerplate by 10x.
- **Mobile signal processing:** Developed an Android-based signal-smoothing algorithm using spherical geometry, acceleration, and last-known-speed data to reduce coordinate jitter in lossy rail-yard and intermodal-trucking environments.

Professional Experience

T-Mobile | Principal Solutions Architect & DevOps Engineer | 2021 to 2026

- Migrated 10+ years of Sprint Jira Data Center metadata and device certification data to Jira Cloud in 3 months with zero downtime for 3,000+ enterprise users, powered by a custom PHP hydration engine that unified remote JSON and local ORM data.
- Built automation tooling with Node.js SDKs and Node-RED nodes integrated with Ansible to capture playbook errors and generate structured Jira incidents automatically.
- Automated fleet mobilization across 132 Spirent Landslide traffic-generation VMs by programmatically updating Netplan configurations and staging loops to prevent host lockouts.
- Reduced a vulnerability backlog of 100+ monthly exploits across 97 hosts by building an idempotent Ansible patch management framework for enterprise security agents.
- Delivered a serverless Angular Universal and Express-based 5G Speed Calculator for the network team, using AWS Lambda microservices to reduce infrastructure overhead and accelerate deployment.
- Architected the Branded Experience Portal monorepo and delivered reusable Angular components for distribution as independent corporate NPM packages.

Sprint | Principal Engineer & Application Developer III | 2017 to 2021

- Architected Porcupine, a RabbitMQ-based event bus that streamlined workflow-driven notifications from Jira to connected tools, improving status visibility across systems.
- Automated a 10-year-old enterprise SKU bottleneck within 6 months by designing a human-in-the-loop pipeline and building the full frontend for the SKU Portal.
- Integrated the SKU Portal with Porcupine to support multi-stage TPM and reviewer approval workflows.
- Architected a data-agnostic PHP framework with a custom hydration engine, unifying SQL, NoSQL, JSON, and XML into a single entity-managed ORM and reducing boilerplate by 10x.
- Validated a Dremio lakehouse proof of concept across MS SQL, MySQL, and MongoDB, demonstrating zero-ETL query federation across relational and NoSQL systems.
- Built observability across 45 VMs using Docker consumers, .NET Core heartbeat telemetry, ELK, Prometheus, Grafana, and Rocket.Chat alerts.
- Served as principal technical advisor to leadership, with final sign-off authority on high-risk deployments and responsibility for growing the department infrastructure budget from \$300K to \$1.3M.

Sprint | Product Development Engineer II | 2013 to 2017

- Built a drag-and-drop prototyping platform for Out-of-Box Experience (OOBE) simulation with validation engines that captured CPU, RAM, and battery telemetry from mobile devices.
- Inherited the Device Development and Certification requirements database and hardened it for production by introducing a component-based jQuery architecture, strict database

transactions for proprietary import/export workflows, and Jira-driven publishing lifecycle controls.

- Led weekly engineering reviews with retail testing leadership to resolve system anomalies and define release timelines for the SLATE ecosystem.

EXPERIS (Contractor at Sprint) | Technical Support Rep II | 2012 to 2013

- Wrote and maintained XML execution scripts for the automated SLATE diagnostic framework used across Sprint's national retail footprint.
- Improved legacy device performance tracking applications to support automated on-site device validation at in-store service centers.

MO-KAN Container Services | System Administrator | 2011 to 2012

- Migrated failing HP G3 bare-metal systems running Windows Server 2003 to a modern high-availability architecture.
- Designed and deployed a redundant VMware environment with iSCSI SAN and Cisco switching through a vendor partnership.
- Took operational ownership of the new infrastructure and reduced enterprise downtime by 90% across 3 distributed locations.
- Managed disaster recovery, failover automation, network security, and centralized domain architecture.

Additional Professional Experience

Trucking Central | Software Consultant (Contract)

- Developed early Android-based GPS and telematics applications for real-time intermodal logistics tracking.

Innovative Solutions | Mobile Solutions Technician

- Deployed and optimized Nextel communication devices and mobile diagnostic tools for municipal first responders and commercial operations.

Technical Environment Matrix

- **Patents & Core Innovation:** US Patent #10,366,069 (Systems and Methods for Database Management and Administration)
- **Architectural Disciplines:** Systems Architecture, Distributed Systems, Microservices, API Design, Platform Abstractions, Developer Experience (DevEx), Internal Tooling Design, SRE, DevSecOps, Enterprise Cloud Migration
- **Infrastructure & Automation:** Ansible, Ansible Automation Platform (AAP), Infrastructure as Code (IaC), CI/CD Pipelines, Node-RED, Core XML Scripting, Serverless, VMware vSphere/ESXi, iSCSI SAN Architecture

- **Data Engineering & Distributed Ecosystems:** RabbitMQ (Porcupine Message Bus), Dremio (Lakehouse Federation), Zero-ETL Query Federation, ELK Stack, Prometheus, Grafana, PostgreSQL, MS SQL, MongoDB, Rocket.Chat Integration
- **Backend Environments & Languages:** Node.js, .NET Core (2.x), C#, PHP, Native Modules, REST APIs, Custom SDK Development, Zero-Dependency Design, Custom ORM Design
- **Frontend Web Frameworks:** Angular, Vue.js 2, TypeScript, JavaScript (ES6+), Client-Side Architecture, UI/UX Performance Optimization

Community Impact & Leadership

Seek and Serve (501c3) | Technical Advisory Board Member | 2026 to Present

- Provide architectural oversight and technical logistics strategy to support regional community outreach expansion.

33 Meals | Technical Advisor & Infrastructure Volunteer | 2026 to Present

- Built the digital architecture and logistics pipeline for the nonprofit subsidiary to support regional food insecurity efforts.
- Established core system deployments, secure domain configurations, and compliance planning for distribution operations.
- Designed a data-driven delivery framework using enterprise architecture principles to scale humanitarian food delivery.

Education

- **Associate of Arts (AA), Graphic Design** – Johnson County Community College